0 / EMPTY
MAIN20 + feed delay

→

10 / INFEED
Conveyor → test

→

20 / TEST
MAIN90 + convey

→

30 / OUTFLOW
Conveyor → outfeed

→

40 / WAIT PICKUP
MAIN10 + home
Then model0

SETTINGS / 100ms COUNTS		
Setting	Simulated function	Counts
D01210	Feed delay	5
D01211	Intake travel time	20
D01212	Infeed clear time	2
D01213	Outfeed travel time	15
D01214	Pickup delay	10
D01215	Clamp response time	4
D01216	Gate arrival time	5
D01217	Gate home time	5

FAULT INJECTION /1 ENABLED	
Injected input	Failure
M02010	No test position
M02011	Clamp will not close
M02012	Clamp will not open
M02013	Selected gate will not arrive
M02014	No outfeed confirmation
M02015	Gate will not return home

SCAN AND MEMORY

SIM_IO executes before MAIN; each program has END.
D01200 current / D01201 next / commit once.
TMR travel retains partial travel on motion OFF.
Unsafe/STOP freezes posture; clear only by accepted M02003.
SIM_IO writes M01000/M01004..0B; M01001..03 are commands, M0100C is receiver-owned.

Main reaches10 while product remains40: pickup timer T0059 completes before next continuous start. Product is not deleted by process RESET alone.

Current control logic and model are separate. Physical I/O replacement changes the input/output mapping, not the plant simulation into a real machine.

NO.	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG TITLE
SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell
SIM_IO PLANT RESPONSE AND FAULT INPUTS

DRAWN BY	DATE
J. H. KO	2026-10-14

DWG NO
015

SHEET NO
1 OF 1